

EDWARD PRASCHIFKA

Graduate Software Engineer

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Graduate software engineer with experience building tested software in C/C++ and Python across embedded systems, backend development and machine learning, including an embedded software internship at Silentium Defence. Built a C++ vector library, a Flask-based graph explorer supporting 2,000+ nodes, and a neural network from first principles. Strong academic results, with HSK 5 Mandarin and a year of study in China through the ANU CIW China Studies Fellowship.

SKILLS

Languages: Python, C, C++, JavaScript, HTML, CSS

Backend: Flask, REST APIs, WebSockets

Testing: PyTest, Catch2, test-driven development

Tooling: Git, Linux, CMake, Jira

EXPERIENCE

Embedded Software Engineer Intern — Silentium Defence, Adelaide *Aug 2024 – Dec 2024*

- Completed a report comparing a commercial drone's built-in API against a specialised GPS system for streaming real-time drone location data in an internet-denied environment; recommended the GPS system because it works with any drone, avoiding lock-in to one drone model.
- Installed LineageOS (Android) on a Raspberry Pi 5 to run the GPS system's Android-only software and receive its location data.
- Implemented a bidirectional C WebSocket connection between a Raspberry Pi 5 and client laptop, enabling real-time drone-location streaming.
- Wrote a 30-page illustrated setup guide with step-by-step instructions and troubleshooting so teammates could reproduce the system.

PROJECTS

TinyVector *Sep 2026* | github.com/edwardpraschifka/tiny-vector

- Built a from-scratch, templated C++ vector class implementing the full Rule of Five with self-assignment protection, const-correct accessors, and both bounds-checked and unchecked element access.
- Developed a Catch2 test suite covering construction, copy/move semantics, and arithmetic operators, validated under AddressSanitizer to catch memory errors at the point of occurrence.

Orbit *Aug 2026* | github.com/edwardpraschifka/orbit | [Live demo](#)

- Built an interactive graph explorer for Steam friend networks: users initialize the graph with their own Steam profile, and click nodes to expand each user's friends into the graph.
- Integrated three Steam Web API endpoints (GetFriendList, GetPlayerSummaries, GetOwnedGames) through a Flask backend; right-clicking a node opens a sidebar of that user's per-game playtime statistics.
- Cached player summaries and friend lists by Steam ID, cutting cached expansion time from ~2 s to ~100 ms; the graph supports 2,000+ nodes.

MLP From Scratch *May 2026 – Jul 2026* | github.com/edwardpraschifka/mlp-from-scratch

- Implemented a feedforward neural network from first principles in NumPy: forward propagation, backpropagation, and gradient descent without high-level ML frameworks.
- Developed a PyTest test suite to test the model's forward propagation, backpropagation, and gradient descent.
- Achieved >97% accuracy on the MNIST dataset.

EDUCATION

Australian National University CIW China Studies Fellowship

Xi'an Jiaotong Liverpool University, Suzhou, China *Feb 2025 – Feb 2026*

A year-long study program focused on Chinese language, culture, history and philosophy.

Bachelor of Mathematical & Computer Sciences and Diploma in Languages (Chinese Studies)

The University of Adelaide *Jan 2022 – Dec 2024*

GPA: 6.08/7 (Bachelor), 6.5/7 (Diploma)

Relevant coursework: Computer Systems (HD), Software Engineering & Project (HD), Object Oriented Programming (HD), Algorithm & Data Structure Analysis (D), Applied Probability III (D), Using Machine Learning Tools (D)

LANGUAGES

- Mandarin — HSK 5 (2,500+ vocabulary), government-standard proficiency exam.